10/08/2026 T-6. (L2- Ridge Regression).

Q. ~~Study~~

Bedrooms	Age (Years)	Price ($ 10000s)
3	5	2
3	3	3

Sol:-

Bedroom (X_1)

Age (X_2)

Price (Y)

$$Y = m_1 X_1 + m_2 X_2 + C.$$

$$2 = C + 3m_1 + 5m_2$$

$$3 = C + 3m_1 + 3m_2$$

$$\underbrace{\begin{pmatrix} 2 \\ 3 \end{pmatrix}}_{Y} = \underbrace{\begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} C \\ m_1 \\ m_2 \end{pmatrix}}_{W}$$

$$W = (X^T X + \lambda I)^{-1} \cdot (X^T Y)$$

Take $\lambda = 1$; $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$W = \left[\begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix} + 1 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right]^{-1} * \left[\begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right]$$

$$= \left[\begin{pmatrix} 2 & 6 & 13 \\ 6 & 18 & 39 \\ 13 & 39 & 89 \end{pmatrix} + 1 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right]^{-1} * \left[\begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right]$$

$$= \begin{bmatrix} 2 & 6 & 13 \\ 6 & 19 & 39 \\ 13 & 39 & 90 \end{bmatrix}^{-1} * \left[\right.$$

Quick Work

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M T W T F S S

$$= \left(\begin{bmatrix} 2 & 6 & 8 \\ 6 & 18 & 24 \\ 8 & 24 & 34 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 15 \\ 19 \end{bmatrix}$$

$$|A| = \begin{bmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 15 \\ 19 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{vmatrix}$$

$$= 3(19 \times 35 - 24 \times 24) - 6(6 \times 35 - 24 \times 8) + 8(6 \times 24 - 19 \times 8)$$

$$= 3(665 - 576) - 6(210 - 192) + 8(144 - 152)$$

$$= 3(89) - 6(18) + 8(-8)$$

$$= 267 - 108 - 64$$

$$= 95$$